

# As easy as CHNOPS: A theory of biological structure

Abstract: What is life? According to NASA’s working definition, life is ‘a self-sustaining chemical system capable of Darwinian evolution’. Yet another recent influential approach defines ‘lyfe’ as any physical system that jointly satisfies the conditions of dissipation, autocatalysis, homeostasis, and learning, with life as the Earth-specific instance of lyfe. Notwithstanding theoretical progress on these and other related fronts, we contend that no existing account has derived the specific elemental composition of life from a minimal set of formally stated, thermodynamically grounded conditions. In this article, we propose a theory of biological structure that is grounded in conditions C1–C4: metastable origin, dissipative persistence, templated replication, and polymer expressivity. This theory allows us to derive the organic alphabet  $\mathcal{E} = \{\text{C, H, N, O, P, S}\}$  as a minimal consistent solution relative to C1–C4 under a set of Earth-specific, biochemically plausible assumptions A1–A8, a functional partition on  $\mathcal{E} - \{\text{C, H, O}\}$  as the primary energy landscape,  $\{\text{N, P}\}$  as the governance layer, and  $\{\text{S}\}$  as the chemical bridge or interface – from C1–C4, and the formal tetrachotomy of metabolite function – substrate, product, cofactor or carrier, and enzyme – from our functional partition on  $\mathcal{E}$ .

Keywords: Life; biological structure; organic alphabet; functional partition; metabolite

## §1 Introduction

What is life? According to NASA’s non-binding but still helpful working definition, life is ‘a self-sustaining chemical system capable of Darwinian evolution’ (Brennan, 2023; Hayes, 2015; Mix, 2026). Though popular among scientists, especially in the astrobiology community, at least some philosophers and biologists have expressed concerns about its lack of analytical rigour and limited applicability (Cleland, 2012; Machery, 2012). Another recent influential approach defines ‘lyfe’ as any physical system that jointly satisfies the conditions of dissipation, autocatalysis, homeostasis, and learning, with life as the Earth-specific instance of lyfe that uses an organometallic molecular toolbox to record information about its environment and achieve dynamical order (Bartlett and Wong, 2020). Yet other approaches have defined life as biological systems information-theoretically in terms of their access to information – genetic, epigenetic, metabolic, and morphological – and acquisition of this information over successive cycles of replication and propagation via evolutionary selection (Marshall et al., 2022).

Despite the theoretical progress, no existing account, to our knowledge, has derived the specific elemental composition of life from a minimal set of formally stated, thermodynamically grounded conditions. Our claims to novelty with respect to our theory of biological structure comprise the following:

(Connection to our minimal conditional theorem of life in § 2) Our theory of biological structure is directly connected to our mini-

mal conditional theorem of life (defended elsewhere). According to this theorem, any living system satisfying four conditions –metastable origin (C1), dissipative persistence (C2), templated replication (C3), and polymer expressivity (C4) – must have a minimal genetic alphabet size  $q = 4$  and a codon length  $n \geq 3$ ;

(Derivation of the organic alphabet  $\mathcal{E}$  in § 3) Most of the literature takes the organic alphabet  $\mathcal{E} = \{\text{C, H, N, O, P, S}\}$  as a given, motivated empirically. By contrast, our theory derives this alphabet  $\mathcal{E}$  as a minimal consistent solution relative to C1–C4 under a set of Earth-specific, biochemically plausible assumptions A1–A8. In particular, we use the kinetic window  $\mathcal{W}$  (§ 3.1) and the thermodynamic coupling window  $\mathcal{T}$  (§ 3.4) as formal selection criteria;

(Functional partition on  $\mathcal{E}$  in § 4) Our functional partition on  $\mathcal{E} - \{\text{C, H, O}\}$  as the primary energy landscape,  $\{\text{N, P}\}$  as the governance layer, and  $\{\text{S}\}$  as the chemical bridge or interface – is derived from C1–C4. C1–C2, governing energy storage and dissipative persistence, select for elements whose characteristic modes support the storage and release of free energy and yield the primary energy landscape. C3–C4, governing template replication and polymer expressivity, select for elements whose characteristic modes support information encoding, conformational control, and the transfer of free energy and yield the governance layer;

(Metabolite tetrachotomy as derivation rather than taxonomy in § 5) The formal tetrachotomy of metabolite function – substrate, product, cofactor or carrier, and enzyme – is derived rather than stipulated: each metabolite is the molecular instantiation of the

elemental architecture of  $\mathcal{E}$  and can be explained in terms of our functional partition on  $\mathcal{E}$ .

## §2 Four conditions for life

According to our theory of chemical structure, the alphabet  $\mathcal{A} = \{a_1, \dots, a_n\}$  is first defined in terms of a finite set of atoms: the 118 chemical element types that are currently known, to be precise. The set of molecules comprises the free commutative monoid on  $\mathcal{A}$ , whereas the set of assemblies of molecules comprises the free commutative monoid on  $\mathcal{M}$ . Deep syntax allows us to distinguish between isomers – molecules with the same molecular formula but different arrangements of atoms — in terms of their constitutional structure (the bonding graph  $G_\alpha$ ) and spatial structure (the geometry  $\xi$ ). Where  $\mathcal{A}$  denotes the set of all assemblies of molecules,  $\mathcal{K}$  denotes the set of allowed contexts (patterns that can be matched),  $\mathcal{P}$  denotes the set of reaction rules (allowed transformations on matched patterns),  $(\mathcal{A}, \mathcal{K}, \mathcal{P})$  yields the full syntactic data of the system. Furthermore,  $\mathcal{S}(\alpha_0, \mathcal{P})$  denotes the set of all assemblies reachable from the initial state  $\alpha_0$  by a finite sequence of rule applications. This also represents the closure of  $\alpha_0$  under the dynamics generated by the rule set  $\mathcal{P}$ .

Semantics gives physical meaning to our formal syntax  $(\mathcal{A}, \mathcal{K}, \mathcal{P})$  by evaluating a system under two external thermodynamic parameters – external energy flux  $\mathcal{F}$  and entropy production rate  $\dot{S}_{\text{prod}}$  – and an internal parameter  $B(\alpha)$ .  $B(\alpha)$ , denoting the balance of production versus consumption within a system  $\alpha$ , can be computed as  $\frac{P(\alpha)}{C(\alpha)} \in \mathbb{R}^{\geq 0}$ , where  $P(\alpha)$  denotes the rate at which  $\alpha$  produces internal resources and  $C(\alpha)$  denotes the rate at which  $\alpha$  consumes internal resources. An interpretation  $\mathcal{I}$  then assigns each system  $\alpha$  to one of four thermodynamic regimes (stable, metastable,

dissipative, runaway). In addition, a living chemical system is an assembly  $\alpha \in \mathcal{A}$  satisfying the following conditions:

- (C1 or metastable origin)  $\exists \alpha_0 (\mathcal{S}(\alpha_0) = B), \alpha_0 \xrightarrow{\mathcal{P}_{\text{trans}}^*} \alpha$  for  $\mathcal{P}_{\text{trans}} \subset \mathcal{P}$ ;
- (C2 or dissipative persistence)  $\mathcal{S}(\alpha) = C$   
 $\therefore F \gtrsim F_{\text{high}}; \dot{S}_{\text{moderate}} \lesssim \dot{S}_{\text{prod}} \lesssim \dot{S}_{\text{high}}; B(\alpha) > 1$ ;
- (C3 templated replication)  $\exists \alpha_r \subseteq \alpha, \mathcal{P}_r \subset \mathcal{P}$  with error  $\epsilon$ , where  $\epsilon$  denotes the error parameter (per-step error rate). Certain replication rules  $\mathcal{P}_r$  allow structures to be copied using an existing strand as a template, producing a complementary strand that is approximately identical to the original, up to errors occurring with probability  $\epsilon$  per replication step;
- (C4 or polymer expressivity) Where  $\mathcal{S}(\alpha_0, \mathcal{P})$  denotes the set of all assemblies reachable from the initial state  $\alpha_0$  or closure under the rule set  $\mathcal{P}$ ,  $\mathcal{S}(\alpha_0, \mathcal{P})$  contains bonding graphs  $G_\alpha$  with unbounded depth and  $\geq 20$   $\xi$ -isomorphic monomers.

C1 reminds us that life emerges from the transition between metastable and dissipative regimes ( $B \rightarrow C$ ). C2 ensures that each living system  $\alpha$  is actively maintained under flux and not merely stable or metastable. C3 presupposes template-based copying as the mechanism of heritable replication: certain replication rules  $\mathcal{P}_r \subset \mathcal{P}$  allow structures to be copied using an existing strand as a template, producing a complementary strand that is approximately identical to the original. C4 requires polymer expressivity, not merely replication. The reachable assembly space  $\mathcal{S}(\alpha_0, \mathcal{P})$  has to include bonding graphs  $G_\alpha$  with unbounded depth and  $\geq 20$   $\xi$ -isomorphic monomers. C4 marks the transition from systems with a limited, fixed repertoire of structures to open-ended systems capable of producing the diverse

range of life-essential functional components (for instance, enzymes, membranes, and so on).

### §3 {C, H, N, O, P, S} as a minimal consistent solution

A restricted number of Earth-specific, biochemically plausible assumptions will allow us to derive the organic alphabet  $\mathcal{E} = \{\text{C, H, N, O, P, S}\} \subsetneq \mathcal{A}$  as a minimal consistent solution relative to C1–C4 and these assumptions:<sup>1</sup>

- (A1) The biologically relevant temperature range for an aqueous context and water-based chemistry is  $T \approx 300$  K;
- (A2) Water is assumed to be the bulk medium for life-relevant chemical reactions;
- (A3) Chemical reaction rates obey Arrhenius kinetics  $k = Ae^{-E_a/k_B T}$ , where  $A$  denotes the pre-exponential factor and  $E_a$  denotes the activation energy;
- (A4) The geological timescale is  $t_{\text{geo}} \sim 10^9$  years  $\approx 10^9 \times 3.15 \times 10^7$  or  $3.15 \times 10^{16}$  seconds;
- (A5) The enzymatic barrier reduction (catalytic boost) is 60–200 kJ/mol;
- (A6) Chemical properties – bond dissociation energies, electronegativity, and so on – can be derived from the standard periodic table and the known periodic table chemistry. Our subproof does not claim to derive atomic properties from first principles;
- (A7) A living assembly  $\alpha$  is modelled as a dissipative structure with  $B(\alpha) > 1$ . Descriptions of the regimes A, B, C, and D and

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<sup>1</sup> If we have a different context (for instance, a different solvent, a different temperature, a different geological baseline, and so on), the numerical thresholds might shift, but the overall logical structure of the proof would remain largely intact.

the conditions C1–C4 to be satisfied can be found in § 2;

(A8) The environment is a single-solvent and single-temperature one.

### 3.1 Carbon

C1 requires  $\alpha_0$  to be metastable in regime B. Its covalent bonds must resist spontaneous rearrangement at ambient temperature, which formally requires that  $\Delta E^\ddagger \gg k_B T$ , where  $k_B$  denotes the Boltzmann constant and  $T \approx 300$  K (A1). C4 requires polymer expressivity: the reachable assembly space  $\mathcal{S}(\alpha_0, \mathcal{P})$  contains bonding graphs  $G_\alpha$  with unbounded depth and  $\geq 20$   $\xi$ -isomorphic monomers. This formally requires a valence-4 atom as the skeletal element, since valence-4 is the minimal threshold valence for building structures of unbounded topological complexity such as chains, rings, branches, and three-dimensional scaffolds.<sup>2</sup> Together, C1 and C4 impose a kinetic window  $\mathcal{W}$ :  $150 \text{ kJ/mol} \ll E_{\text{bond}} \lesssim 350 \text{ kJ/mol}$ .<sup>3</sup> Among the valence-4 atoms,

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<sup>2</sup> Valence refers to the number of bonds an atom can form. A valence- $n$  atom can form  $n$  bonds simultaneously. When  $n < 4$ , the only possible structures that can be formed are either terminal structures with no branching (valence-1 atoms like H), linear chains with limited branching (valence-2 atoms like O), or planar networks (valence-3 atoms like N).

<sup>3</sup> At temperature  $T \approx 300$  K,  $k_B T$  can be computed as  $k_B T \approx 1.380 \times 10^{-23} \times 300 \approx 4.14 \times 10^{-21}$  J. Multiplying this value by Avogadro’s constant ( $N_A$ ) will yield  $4.14 \times 10^{-21} \times 6.022 \times 10^{23} \div 10^3$  kJ/mol or 2.5 kJ/mol. Since  $E_{\text{bond}} \gg k_B T$  and the empirical floor for covalent bond dissociation energies is 150 kJ/mol, we can take the lower bound as  $E_{\text{bond}} \gg 150$  kJ/mol. For the dissipative regime C, the  $B \rightarrow C$  transition must be observable. Using the Arrhenius equation with the minimum observable rate constant over geological time (A3–A4):  $t_{\text{geo}} \sim 10^9$  years  $\approx 10^9 \times 3.15 \times 10^7$  or  $3.15 \times 10^{16}$  seconds;  $k_{\text{min}} \sim \frac{1}{t_{\text{geo}}} = \frac{1}{3.15 \times 10^{16}}$  or  $3.3 \times 10^{-17} \text{ s}^{-1}$  (minimum rate constant); and  $\Delta E_{\text{geo}}^\ddagger$  (kinetic ceiling)  $= \Delta E^\ddagger = RT \ln \frac{k_B/h}{k_{\text{min}}}$ , where  $R = 8.314 \text{ Jmol}^{-1}\text{K}^{-1}$  represents the gas constant,  $T = 300$  K represents the room temperature (A1),  $k_B = 1.380 \times 10^{-23} \text{ J/K}$  represents the Boltzmann constant, and  $h = 6.626 \times 10^{-24} \text{ J-s}$  represents Planck’s constant.  $\Delta E_{\text{geo}}^\ddagger$  can be computed as approximately 150 kJ/mol. Adding the enzymatic

carbon and silicon are the most chemically effective candidates for C4-style polymer expressivity. However:

C — C bond: 346 kJ/mol  $\in \mathcal{W}$

Si — O bond: 452 kJ/mol  $\notin \mathcal{W}$

$\therefore \text{C} \in \mathcal{E}, \text{Si} \notin \mathcal{E}$

Silicon’s preferred bonding mode drives it outside the kinetic window  $\mathcal{W}$  into the stable regime A, violating C1. Therefore, carbon is the unique valence-4 element satisfying both C1 and C4 under A1–A8.

### 3.2 Hydrogen

C2 requires dissipative persistence:  $B(\alpha) > 1$  must be sustained under high flux  $F \gtrsim F_{\text{high}}$ . Under aqueous conditions (A2), sustaining  $B(\alpha) > 1$  requires the selective channelling of flux into productive rather than degradative transformations. This selective channelling is achieved via hydrogen bonding and proton transfer. H is the only element that can be transferred as a bare proton ( $\text{H}^+$ ), with no electron shell to impede the transfer. This satisfies the proton-transfer requirement of C2 uniquely:

$\therefore \text{H} \in \mathcal{E}$

### 3.3 Nitrogen and oxygen

The productive channelling of flux required by C2 further requires atoms that can act simultaneously as hydrogen-bond donors and acceptors and

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barrier reduction of 60–200 kJ/mol (A5) raises the effective ceiling to  $\Delta E_{\text{max}}^{\ddagger} \approx 350$  kJ/mol (upper bound).

participate in acid-base chemistry. This formally selects for light, electronegative, period-2 elements. Among the period-2 elements, at least some are too electronegative (F, a poor H-bond donor), others are chemically inert and excluded from covalent chemistry by definition (Ne, a noble gas), yet others are electropositive (Li and Be), and still others are electron-poor (B). This leaves O and N as the unique period-2 elements satisfying the hydrogen-bonding and acid-base requirements of C2:

$$\therefore \{N, O\} \in \mathcal{E}$$

### 3.4 Phosphorus and sulphur

C1 requires that free energy is stored in a metastable form (regime B) and released in coupled quanta under dissipative conditions (regime C). This formally imposes a thermodynamic coupling window  $\mathcal{T}$ :

$$(\mathcal{T} \text{ or thermodynamic coupling window}) \Delta G_{\text{hydro}} \approx 20\text{--}60 \text{ kJ/mol} \\ \text{per hydrolysis event}$$

$\mathcal{T}$  is bounded below by the minimum free energy required to drive the smallest biosynthetic reactions (20 kJ/mol) and above by the maximum usable quantum of free energy that biological machinery can transduce efficiently (60 kJ/mol). Two bond types satisfy  $\mathcal{T}$  under aqueous conditions (A2): phosphoanhydride bonds ( $\approx 30\text{--}60 \text{ kJ/mol} \in \mathcal{T}$ ) and thioester bonds ( $\approx 30 \text{ kJ/mol} \in \mathcal{T}$ ) (Fig. 1).

Under A1–C8, no other element forms bonds lying within  $\mathcal{T}$  at the B  $\rightarrow$  C interface required by C1:



Figure 1: From left to right: a phosphoanhydride bond (30–60 kJ/mol) and a thioester bond (30 kJ/mol). For the thioester bond, R is essentially a placeholder for the rest of the molecule

Phosphoanhydride bonds:  $\Delta G_{\text{hydro}} \approx 30\text{--}60 \text{ kJ/mol} \in \mathcal{T}$ ;

Thioester bonds:  $\Delta G_{\text{hydro}} \approx 30 \text{ kJ/mol} \in \mathcal{T}$ ;

$\therefore \{\text{P}, \text{S}\} \in \mathcal{E}$

It follows that  $\mathcal{E} = \{\text{C}, \text{H}, \text{N}, \text{O}, \text{P}, \text{S}\}$  is the minimal consistent alphabet satisfying C1–C4 under A1–A8.

#### §4 Functional partition on $\mathcal{E}$

The derivation of  $\mathcal{E}$  in § 3 was driven by four formal conditions C1–C4. These exact conditions impose a functional partition on  $\mathcal{E}$  itself. C1–C2, governing energy storage and dissipative persistence, select for elements whose characteristic modes support the storage and release of free energy. This yields the primary energy landscape  $\{\text{C}, \text{H}, \text{O}\}$ . C3–C4, governing template replication and polymer expressivity, select for elements whose characteristic modes support information encoding, conformational control, and the transfer of free energy. This yields the governance layer  $\{\text{N}, \text{P}\}$ . S, whose characteristic modes span both the kinetic window  $\mathcal{W}$  and the thermodynamic coupling window  $\mathcal{T}$ , functions as a chemical bridge or interface between the primary energy landscape  $\{\text{C}, \text{H}, \text{O}\}$  and the governance layer  $\{\text{N}, \text{P}\}$ .

#### 4.1 Primary energy landscape

The primary energy landscape  $\{\text{C}, \text{H}, \text{O}\}$  is therefore a landscape of the reduction state: the degree to which carbon skeletons carry an excess of hydrogen over oxygen. C1–C2 jointly require that free energy is stored in reduced carbon skeletons and released by their oxidation. The hydrogen-to-oxygen ratio of a molecule is therefore a direct measure of its position in the primary energy landscape:

H:O ratio of glucose ( $\text{C}_6\text{H}_{12}\text{O}_6$ ): 2:1

H:O ratio of pyruvic acid ( $\text{C}_3\text{H}_4\text{O}_3$ ): 4:3

$\therefore \text{C}_6\text{H}_{12}\text{O}_6 \succ \text{C}_3\text{H}_4\text{O}_3$ , where  $\succ$  denotes an ordered relation in the primary energy landscape

C — H bonds characterize reduced, energy-rich carbon, whereas C = O bonds characterize oxidized, energy-poor carbon. Energy is released as C — H bonds are converted into C = O bonds. Catabolism and anabolism, in turn, are two opposite ways of moving through this energy landscape. Catabolism traverses the landscape downhill, oxidizing carbon skeletons and harvesting free energy. Anabolism runs uphill, using the harvested energy to reduce simple, oxidized precursors into complex, hydrogen-rich molecules.

#### 4.2 Governance layer

The governance layer  $\{\text{N}, \text{P}\}$  is the layer of control. C3 requires templated replication, while C4 requires polymer expressivity with  $\geq 20$   $\xi$ -isomorphic monomers. Both C3–C4 formally require elements capable of encoding information and enforcing conformational specificity. Nitrogen satisfies these requirements through its lone pair chemistry. Each nitrogen atom carries a

lone pair of electrons that can behave either as a base (grabbing a proton and forming a bond with an electron-poor species) or as a nucleophile (donating an electron pair to form a new covalent bond). When the lone pair is localized, as in a free amine, the nitrogen is pyramidal, the C—N bond rotates freely, and the carbon scaffold remains conformationally flexible. However, when the lone pair is delocalized into a neighbouring carbonyl (C=O) group, as in an amide, the C—N bonds acquire a partial double-bond character, rotation is restricted, and the atoms will be constrained to an approximately planar arrangement.

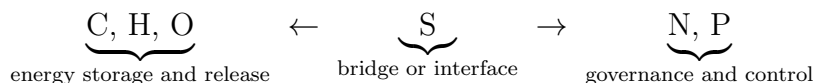
This electronic delocalization introduces conformational rigidity into the peptide backbone and drastically reduces the range of conformations that a polypeptide chain can explore. The constrained geometry is what makes amide bonds, rather than the free amine, the structural unit of proteins: reduced degrees of conformational freedom in the sequence-to-structure mapping make protein folding more reproducible. Beyond catalysis, nitrogen forms the backbone of nucleobases (satisfying C3 or templated replication) and of the peptide bond (satisfying C4 or polymer expressivity). Phosphorus chiefly appears as phosphate. Its tetrahedral, highly charged geometry makes phosphate groups ideal for energy transfer: ATP (adenosine triphosphate), GTP (guanosine triphosphate), and related nucleotides act as portable energy currency. This makes chemical energy movable and controllable, rather than merely stored or released (§ 4.1).

### 4.3 Bridge or interface

The chemical bridge or interface {S} occupies a position that neither the primary energy landscape {C, H, O} (§ 4.1) nor the governance layer {N, P} (§ 4.2) alone can fill. While oxygen and sulphur are in the same group of the

periodic table, they solve different chemical problems. Oxygen’s electrons are held tightly close to the nucleus, making it a sluggish nucleophile. By contrast, sulphur’s electron cloud is large and diffuse, allowing it to carry out fast, powerful nucleophilic attacks and form new bonds rapidly. This large and diffuse electron cloud is easy to distort and easy to share: sulphur is therefore an excellent actor (conducting nucleophilic attacks with unmatched speed, binding strongly to soft metals such as iron, switching redox states reversibly) and a poor stabilizer (unable to lock away energy in a stable ester the way oxygen does).

Sulphur neither stores energy in the manner of  $\{\text{C, H, O}\}$  nor encodes information in the manner of  $\{\text{N, P}\}$ . Instead, as a chemical bridge between energy and control, sulphur transfers and transforms, moving things along and making reactions occur that otherwise would not. More generally, thermodynamics without governance –  $\mathcal{E} \setminus \{\text{N, P, S}\}$  – is mere combustion, whereas governance without thermodynamic grounding –  $\mathcal{E} \setminus \{\text{C, H, O, S}\}$  – is uncoupled from its energy source and cannot be sustained. Life requires both thermodynamic grounding and governance, coupled and inseparable, with S as the chemical bridge or interface:



## §5 Metabolism

Our organic alphabet  $\mathcal{E} = \{\text{C, H, N, O, P, S}\}$  and its functional partition into an energy landscape, a governance layer, and a chemical bridge (§ 4) tell us what chemistry is possible. By contrast, metabolites are the molecular vocabulary through which this chemistry is enacted in living organisms. Each

metabolite  $\alpha \in \mathcal{A}_m$  is not merely a collection of atoms from  $\mathcal{E}$ : it is a chemical structure whose functional identity is determined by what happens to it across a reaction  $\rho \in \mathcal{P}_m$ .

The formal conditions C1–C4 jointly entail a partition of molecular function. C2 requires that  $B(\alpha) > 1$ : a living system must produce internal resources faster than it consumes them ( $P(\alpha) > C(\alpha)$ ). This, in turn, formally requires molecules that are consumed in energy-yielding transformations (substrates whose bonds are broken and atoms redistributed into products), molecules that are produced by those transformations (products), and molecules that mediate these transformations without being consumed (cofactors or carriers). C3–C4 additionally require templated replication and polymer expressivity: these conditions require molecules that are structurally invariant across reactions. These molecules, neither consumed, produced, nor toggled, yet capable of lowering activation barriers with selectivity, are known as enzymes. The tetrachotomy of substrate (consumed), product (produced), cofactor or carrier (toggled), and enzyme (structurally invariant) is therefore neither an arbitrary stipulation nor a taxonomic convenience but rather a necessary consequence of C1–C4. Indeed, any living chemical system satisfying our minimal conditional theorem of life (§ 2) must contain molecules playing each of these four roles.

These four formal roles – products, substrates, cofactors or carriers, and enzymes – recapitulate, at the molecular level, the functional partition of  $\mathcal{E}$  established in § 4. Substrates and products are the molecular currency of the primary energy landscape: reduced carbon skeletons (high H:O ratio), including glucose ( $\text{C}_6\text{H}_{12}\text{O}_6$ ) and fatty acids such as palmitic acid ( $\text{C}_{16}\text{H}_{32}\text{O}_2$ ), enter from above as substrates to be consumed during catabolic reactions, while oxidized endpoints such as  $\text{CO}_2$  and  $\text{H}_2\text{O}$  exit below as products.

| Formal role            | Elemental identity                                                         | Metabolite function                                                                                                                                                                   |
|------------------------|----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Substrate              | Primary energy landscape<br>( $\mathcal{E} \setminus \{\text{N, P, S}\}$ ) | Reduced carbon skeletons<br>( $\text{C}_6\text{H}_{12}\text{O}_6$ , $\text{C}_{16}\text{H}_{32}\text{O}_2$ ) enter<br>the energy landscape from<br>above and are consumed<br>downhill |
| Product                | Primary energy landscape<br>( $\mathcal{E} \setminus \{\text{N, P, S}\}$ ) | Oxidized endpoints ( $\text{CO}_2$ ,<br>$\text{H}_2\text{O}$ ) exit the energy land-<br>scape at the bottom                                                                           |
| Cofactor or<br>carrier | Governance layer<br>( $\mathcal{E} \setminus \{\text{C, H, O, S}\}$ )      | Cofactors or carriers<br>( $\text{NAD}^+/\text{NADH}$ , $\text{ATP}/\text{ADP}$ )<br>shuttle harvested energy<br>across reactions and persist in<br>toggled states                    |
| Enzyme                 | Full organic alphabet ( $\mathcal{E} \setminus \emptyset$ )                | Catalytic agents (nitrogenase,<br>cytochrome c oxidase) couple<br>the energy landscape to the<br>governance layer by lowering<br>activation barriers with selec-<br>tivity            |

Table 1: Biochemical roles of the different metabolites

Cofactors and carriers are the governance layer  $\{N, P\}$  made molecular: they persist across reactions and toggle their internal states.  $NAD^+$  or  $NADH$  shuttles electrons harvested from carbon oxidation, while  $ATP/ADP$  carries the phosphoryl group whose transfer drives endergonic reactions. Enzymes span the full alphabet  $\mathcal{E}$ , and it is their elemental completeness that equips them to couple the energy landscape to the governance layer. As shown in Table 1, the formal tetrachotomy of metabolite function – substrate, product, cofactor or carrier, and enzyme – is not stipulative but derived: it is the molecular instantiation of the elemental architecture of  $\mathcal{E}$ , itself derived from C1–C4.

## §6 Conclusion

In conclusion, we have shown that the organic alphabet  $\mathcal{E} = \{C, H, N, O, P, S\}$  is not some empirical coincidence but a minimal consistent solution to the constraints imposed by C1–C4 under A1–A8. Our formal derivation proceeds in a deductive chain: the kinetic window  $\mathcal{W}$ , the thermodynamic coupling window  $\mathcal{T}$ , and other structural requirements imposed by polymer expressivity and templated replication result in  $\mathcal{E}$  being selected from the full set of 118 known chemical elements or the chemical alphabet  $\mathcal{A}$ . In addition, our functional partition on  $\mathcal{E} - \{C, H, O\}$  as the primary energy landscape,  $\{N, P\}$  as the governance layer, and  $\{S\}$  as the chemical bridge or interface – is not imposed from without but derived from within our theory (C1–C4). Last but not least, the metabolite tetrachotomy of substrate (consumed), product (produced), cofactor or carrier (toggled), and enzyme (structurally invariant) follows as the molecular instantiation of the elemental architecture of  $\mathcal{E}$ . All things considered, our formal theory of biological structure is unified: its definition of life in terms of the joint satisfaction of conditions

C1–C4, the composition of its chemistry in terms of the alphabet  $\mathcal{E}$ , and the organization of its molecular vocabulary (substrate, product, cofactor or carrier, and enzyme).

## References

- Bartlett, Stuart and Michael L. Wong (2020). “Defining lyfe in the universe: From three privileged functions to four pillars”. In: *Life* 10.4, p. 42.
- Brennan, P. (2023). “Life on other planets: What is life & what does it need”. In: *NASA Exoplanet Exploration*.
- Cleland, Carol E. (2012). “Life without definitions”. In: *Synthese* 185.1, pp. 125–144.
- Hayes, Lindsay, ed. (2015). *Astrobiology Strategy 2015*. National Aeronautics & Space Administration.
- Machery, Edouard (2012). “Why I stopped worrying about the definition of life... & why you should as well”. In: *Synthese* 185.1, pp. 145–164.
- Marshall, Stuart M. et al. (2022). “Formalising the pathways to life using assembly spaces”. In: *Entropy* 24.7, p. 884.
- Mix, Lucas J. (2026). “The Origin, Extension, & Future of the “NASA Definition” of Life”. In: *Astrobiology* 26.1, pp. 66–74.